

Generation Interconnection System Impact Study Report

For Queue Project Q15-0458

Hubbell 138 kV
December 30, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0458. The Interconnection Customer proposes a 78.9 MW wind facility interconnecting at the Hubbell 138 kV substation in the Maryland area. The request was received on April 08, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 212 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$827,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hillsboro-Alum Ridge 138 KV L	Linde-Ripple 138 KV L	143.4	184	6.8
Buckskin Substation-Alum Ridge 138 KV	Linde-Ripple 138 KV	124.4	198	6.8
Buckskin-Vernon 138 KV	Linde Substation-Ripple 138 KV L	109.1	216	6.8
Holston Dam-Navy Mole 138 KV L	Hitchland Substation-N Lake 138 KV	100.3	338	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alum Ridge Substation-Hillsboro 138 KV	Rebuild line with higher-capacity conductor	\$4,348,000	6% (\$272,000)
Alum Ridge-Buckskin 138 KV	Rebuild line with higher-capacity conductor	\$5,412,000	6% (\$338,000)
Buckskin-Vernon 138 KV	Reconductor line	\$3,018,000	6% (\$189,000)
Holston Dam-Navy Mole 138 KV L	Reconductor line	\$811,000	3% (\$28,000)
	Total allocated to project		\$827,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.